

NTRK1 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes NTRK1 gene fusions from the msk_impact_50k_2026 study, covering 78 fusion events. 32/78 fusions (41.0%) were in-frame. 64/78 fusions (82.1%) retained the PF07714 domain. Domain retention was tested with Fisher's exact test ($p=0.0482628$, statistically significant at $\alpha=0.05$). No literature benchmark is configured for NTRK1.

1 Results summary

1.1 Fusion partner frequency

Fusion-partner frequency was tabulated across 78 analyzed fusion events, identifying 43 distinct fusion partner genes. The most frequent partner was LMNA, observed in 12 events.

1.2 Domain retention

PF07714 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 30/32 (93.8%) of in-frame fusions retained the domain, $p=0.0482628$ (statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p -value of 0.000999001.

1.3 Domain disruption

Domain-disruption analysis was skipped: NTRK1 has no `disruption_required_domains` configured; domain-disruption analysis was skipped.

1.4 Cutpoint detection

Cutpoint detection scanned 76 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 501 aa (permutation-corrected $p=0.014985$, statistically significant at $\alpha=0.05$). This is -11 aa from the nearest configured domain boundary at 512 aa.

1.5 Corroborating confidence statistics

Corroborating confidence statistics compared `Domain_retention_flags` groups "retained" ($n=64$) and "not_retained" ($n=12$). For `Frame_status`, retained: 30/36 (83.3%, 95% CI 68.1%-92.1%); not_retained: 2/8 (25.0%, 95% CI 7.1%-59.1%). Welch's t -test comparing

`Tumor_variant_count` between groups gave means of 66.3594 vs 43.25, $p=0.200314$ (not statistically significant at $\alpha=0.05$).

1.6 Composite evidence score

Composite evidence score reported: `domain_disruption` sub-score excluded: the supplied result had no applicable p -value (e.g. `disruption_required_domains` not configured for this gene).

1.7 Exon retention

Exon retention reported: No target exon was supplied in `params` or `GeneConfig`.

1.8 Joint partner

Joint partner reported: NTRK1 has no `gene_pair` configured; joint-partner analysis was skipped.

2 Tables

2.1 composite_score composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
LMNA	12	0.15384615384615385	0.30004340774793187		0.851937984496124	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3887286932002887	1
TPM3	12	0.15384615384615385	0.30004340774793187		0.7467054263565891	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3689975885491259	2
MEF2D	1	0.01282051282051282	0.30004340774793187		1.0	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.36360570572265005	3
TRPM8	1	0.01282051282051282	0.30004340774793187		1.0	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.36360570572265005	4
ARHGEF11	2	0.02564102564102564	0.30004340774793187		0.9430232558139535	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.35773025849545864	5
RABGAP1L	1	0.01282051282051282	0.30004340774793187		0.8860465116279069	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3422394266528826	6
VANGL2	1	0.01282051282051282	0.30004340774793187		0.8860465116279069	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3422394266528826	7
ZBTB7B	1	0.01282051282051282	0.30004340774793187		0.8860465116279069	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3422394266528826	8
SCP2	1	0.01282051282051282	0.30004340774793187		0.8651162790697674	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3383150080482314	9
TPR	5	0.0641025641025641	0.30004340774793187		0.7427906976744186	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3346097307673727	10
LTAP1	1	0.01282051282051282	0.30004340774793187		0.8372093023255813	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3330824499086965	11
ATP1A2	1	0.01282051282051282	0.30004340774793187		0.8186046511627907	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.32959407781567324	12
PLEKHA6	3	0.038461538461538464	0.30004340774793187		0.7627906976744185	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.32874434615198816	13
IRF2BP2	3	0.038461538461538464	0.30004340774793187		0.758139534883721	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3278722531287323	14
ANKRD36	1	0.01282051282051282	0.30004340774793187		0.8069767441860465	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.32741384525753375	15
RAB25	1	0.01282051282051282	0.30004340774793187		0.8046511627906977	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3269777987459059	16
IQGAP3	1	0.01282051282051282	0.30004340774793187		0.7976744186046512	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3256696592110221	17
SMYD2	1	0.01282051282051282	0.30004340774793187		0.7767441860465116	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.32174524060637094	18

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty	Components_applicable	Composite_score	Rank
EML4	1	0.01282051282051282	0.30004340774793187		0.7627906976744185	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3191289615366035	19
EPS15	1	0.01282051282051282	0.30004340774793187		0.7627906976744185	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3191289615366035	20
F11R	1	0.01282051282051282	0.30004340774793187		0.7627906976744185	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3191289615366035	21
PEAR1	1	0.01282051282051282	0.30004340774793187		0.7627906976744185	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3191289615366035	22
AFAP1	1	0.01282051282051282	0.30004340774793187		0.7488372093023256	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.31651268246683606	23
DIAPH1	1	0.01282051282051282	0.30004340774793187		0.7488372093023256	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.31651268246683606	24
TRIM63	1	0.01282051282051282	0.30004340774793187		0.7488372093023256	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.31651268246683606	25
STK11	1	0.01282051282051282	0.30004340774793187		0.7325581395348837	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.31346035688544077	26
KIF21B	1	0.01282051282051282	0.30004340774793187		0.7255813953488373	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.312152217350557	27
GON4L	1	0.01282051282051282	0.30004340774793187		0.7162790697674419	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3104080313040454	28
BCAN	2	0.02564102564102564	0.30004340774793187		0.49534883720930234	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.27379130500708654	29
CTRC	2	0.02564102564102564	0.30004340774793187		0.49534883720930234	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.27379130500708654	30
NOS1AP	1	0.01282051282051282	0.30004340774793187		0.5186046511627906	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.27334407781567327	31
NELFCD	1	0.01282051282051282	0.30004340774793187		0.49534883720930234	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.26898361269939425	32
SEMA4A	1	0.01282051282051282	0.30004340774793187		0.49534883720930234	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.26898361269939425	33
COP1	1	0.01282051282051282	0.30004340774793187		0.4534883720930233	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.2611347754900919	34
TARS2	1	0.01282051282051282	0.30004340774793187		0.2697674418604651	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.22668710107148723	35
P2RY8	1	0.01282051282051282	0.30004340774793187		0.2581395348837209	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.2245068685133477	36
DDR2	1	0.01282051282051282	0.30004340774793187			0.6202755879498323	['confidence_certainty', 'domain_retention', 'recurrence']	0.21674548396633853	37
METTL25B	1	0.01282051282051282	0.30004340774793187			0.6202755879498323	['confidence_certainty', 'domain_retention', 'recurrence']	0.21674548396633853	38
SLAMF6	3	0.038461538461538464	0.30004340774793187		0.11395348837209307	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.20708736940780212	39

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty	Components_applicable	Composite_score	Rank
CADM3	1	0.01282051282051282	0.30004340774793187		0.11395348837209307	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.1974719847924175	40
PRCC	1	0.01282051282051282	0.30004340774793187		0.11395348837209307	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.1974719847924175	41
TAF2	1	0.01282051282051282	0.30004340774793187		0.11395348837209307	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.1974719847924175	42
SHPRH	1	0.01282051282051282	0.30004340774793187		0.0	0.6202755879498323	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.17610570572265005	43

2.2 cutpoint_detection cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
71	0	64	1	11	0.0	0.15789473684210525	0.8016323462331666
120	5	59	2	10	0.423728813559322	0.3043106841520683	0.5166827995746912
182	5	59	3	9	0.2542372881355932	0.10748982443269033	0.9686326464637113
187	6	58	3	9	0.3103448275862069	0.14608214986786255	0.83540284831964
284	10	54	5	7	0.25925925925925924	0.05253839246726117	1.2795232195834179
294	11	53	5	7	0.29056603773584905	0.11464025310482523	0.9406628637664377
321	13	51	5	7	0.3568627450980392	0.14136093824402443	0.8496705809825442
343	14	50	5	7	0.392	0.1614921032975089	0.7918487091109843
359	15	49	5	7	0.42857142857142855	0.28166616270038797	0.550265322823117
377	16	48	5	7	0.4666666666666667	0.29450254706111656	0.530910944782467
379	17	47	5	7	0.5063829787234042	0.31247043251825074	0.505191071445305
383	18	46	5	7	0.5478260869565217	0.494013437398657	0.30626123790080845
393	27	37	5	7	1.0216216216216216	1.0	-0.0
399	39	25	5	7	2.184	0.33960073354007925	0.46903138034780045
405	39	25	6	6	1.56	0.5328211446391313	0.27341854881920497
411	40	24	6	6	1.6666666666666667	0.5236156499522178	0.28098738162793063
414	41	23	6	6	1.7826086956521738	0.5184932375861125	0.2852569034951671
417	46	18	7	5	1.8253968253968254	0.49401343739865705	0.3062612379008084
418	48	16	7	5	2.142857142857143	0.29450254706111656	0.530910944782467
423	49	15	7	5	2.3333333333333335	0.28166616270038797	0.550265322823117
443	50	14	7	5	2.5510204081632653	0.1614921032975089	0.7918487091109843
452	61	3	8	4	10.166666666666666	0.010191376831302902	1.9917671398557228
492	62	2	8	4	15.5	0.004800736843751964	2.3186920995794114
501	64	0	9	3		0.0031294452347083927	2.5045326441976177
571	64	0	10	2		0.023157894736842103	1.6353009244666417
616	64	0	11	1		0.15789473684210525	0.8016323462331666

2.3 domain_retention frame_domain_contingency_table

30	34
2	10

2.4 domain_retention domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained
kinase	Protein kinase domain	76	64	3	9	0.1117283950617284

2.5 frequency Partner_gene_counts

Partner_gene	Event_count
AFAP1	1
ANKRD36	1
ARHGEF11	2
ATP1A2	1
BCAN	2
CADM3	1
COP1	1
CTRC	2
DDR2	1
DIAPH1	1
EML4	1
EPS15	1
F11R	1
GON4L	1
IQGAP3	1
IRF2BP2	3
KIF21B	1
LMNA	12
LTAP1	1
MEF2D	1
METTL25B	1
NELFCD	1
NOS1AP	1
P2RY8	1
PEAR1	1
PLEKHA6	3
PRCC	1
RAB25	1
RABGAP1L	1
SCP2	1
SEMA4A	1
SHPRH	1
SLAMF6	3
SMYD2	1
STK11	1
TAF2A	1
TARS2	1
TPM3	12
TPR	5
TRIM63	1
TRPM8	1
VANGL2	1
ZBTB7B	1

2.6 Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint	breakpoint	breakpoint	protein	protein	breakpoint	breakpoint	main_ref	chr	start	end	domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0014859-T01-IM6-9	P-0014859-T01-IM6		CTRC-NTRK1	CTRC	unknown	three_prime	156841592	7	284	True	retained	1.0	False	Protein kinase domain						CTRC (NM_007272) - NTRK1 (NM_002529) rearrangement: c.88:CTRC_c.850+45:NTRK1del	NA	
EVT-P-0029326-T01-IM6-10	P-0029326-T01-IM6		DIAPH1-NTRK1	DIAPH1	in-frame	three_prime	156843860	8	393	True	retained	1.0	False	Protein kinase domain						DIAPH1 (NM_005219) - NTRK1 (NM_002529) rearrangement: t(1;5)(q23.1;q31.3)(chr1:g.156843860::chr	NA	
EVT-P-0059139-T01-IM7-11	P-0059139-T01-IM7		EML4-NTRK1	EML4	in-frame	three_prime	156844256	9	399	True	retained	1.0	False	Protein kinase domain						fusion: t(1;2)(q23.1;p21)(chr1:g.156844256::chr	NA	
EVT-P-0037901-T01-IM6-12	P-0037901-T01-IM6		EPS15-NTRK1	EPS15	in-frame	three_prime	156844212	9	399	True	retained	1.0	False	Protein kinase domain						fusion: in frame {EML4:NTRK1} EPS15 (NM_001981) - NTRK1 (NM_002529) rearrangement: c.2119+1240:EPS15_c.1195+20:NTRK1	NA	
EVT-P-0009855-T01-IM5-13	P-0009855-T01-IM5		GON4L-NTRK1	GON4L	unknown	three_prime	156843711	8	379	False	retained	1.0	False	Protein kinase domain						fusion: in frame {EPS15:NTRK1} GON4L (NM_001037533) - NTRK1 (NM_002529) rearrangement: c.4141:GON4L_c.1137:NTRK1inv	NA	
EVT-P-0050669-T01-IM6-15	P-0050669-T01-IM6		IQGAP3-NTRK1	IQGAP3	unknown	three_prime	156844408	10	414	False	retained	1.0	False	Protein kinase domain						fusion: mid-exon (GON4L-NTRK1) IQGAP3 (NM_178229) - NTRK1 (NM_002529) fusion: c.2531-131:IQGAP3_c.1241:NTRK1inv	NA	
EVT-P-0022925-T01-IM6-16	P-0022925-T01-IM6		LMNA-NTRK1	LMNA	unknown	three_prime	156844398	10	411	False	retained	1.0	False	Protein kinase domain						fusion: mid-exon {IQGAP3:NTRK1} LMNA (NM_170707) - NTRK1 (NM_002529) fusion (LMNA exons 1-3 fused to NTRK1 exons 10 - 17): c.513+1675:LMNA_c.1231:NTRK1del	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_retained	breakpoint_4	breakpoint_5	main_ref	donor_frame	maintained	disrupted_domain	disrupted_domain	description	annotation	effect_on_frame
EVT-P-0003329-T03-IM6-26	P-0003329-T03-IM6		LMNA-NTRK1	LMNA	unknown	three_prime	156845200	12	452	True	retained	1.0	False	Protein kinase domain				LMNA (NM_170707) - NTRK1 (NM_002529) fusion (LMNA exons 1-2 fused to NTRK1 exons 12 - 17): c.495:LMNA_c.1355-112:NTRK1del Protein Fusion: mid-exon {LMNA:NTRK1} LMNA (NM_170707) - NTRK1 (NM_002529) fusion: c.1428:LMNA_c.1355-142:NTRK1del Protein Fusion: mid-exon {LMNA:NTRK1} LMNA (NM_170707) - NTRK1 (NM_002529) fusion (LMNA exon2 fused with NTRK1 exon12): c.495:LMNA_c.1355-112:NTRK1del Protein Fusion: mid-exon {LMNA:NTRK1} LMNA (NM_170707) - NTRK1 (NM_002529) fusion: c.1699-362:LMNA_c.1251+107:NTRK1del Protein Fusion: in frame {LMNA:NTRK1} LMNA (NM_170707) - NTRK1 (NM_002529) fusion (LMNA exon 10 fused with NTRK1 exon 11) : c.1699-362:LMNA_c.1251+107:NTRK1del Protein Fusion: in frame {LMNA:NTRK1}	NA	
EVT-P-0066239-T01-IM7-27	P-0066239-T01-IM7		LMNA-NTRK1	LMNA	unknown	three_prime	156845170	12	452	True	retained	1.0	False	Protein kinase domain				LMNA (NM_170707) - NTRK1 (NM_002529) fusion: c.1428:LMNA_c.1355-142:NTRK1del Protein Fusion: mid-exon {LMNA:NTRK1} LMNA (NM_170707) - NTRK1 (NM_002529) fusion (LMNA exon2 fused with NTRK1 exon12): c.495:LMNA_c.1355-112:NTRK1del Protein Fusion: mid-exon {LMNA:NTRK1} LMNA (NM_170707) - NTRK1 (NM_002529) fusion: c.1699-362:LMNA_c.1251+107:NTRK1del Protein Fusion: in frame {LMNA:NTRK1} LMNA (NM_170707) - NTRK1 (NM_002529) fusion (LMNA exon 10 fused with NTRK1 exon 11) : c.1699-362:LMNA_c.1251+107:NTRK1del Protein Fusion: in frame {LMNA:NTRK1}	NA	
EVT-P-0003329-T02-IM6-28	P-0003329-T02-IM6		LMNA-NTRK1	LMNA	unknown	three_prime	156845200	12	452	True	retained	1.0	False	Protein kinase domain				LMNA (NM_170707) - NTRK1 (NM_002529) fusion (LMNA exon2 fused with NTRK1 exon12): c.495:LMNA_c.1355-112:NTRK1del Protein Fusion: mid-exon {LMNA:NTRK1} LMNA (NM_170707) - NTRK1 (NM_002529) fusion: c.1699-362:LMNA_c.1251+107:NTRK1del Protein Fusion: in frame {LMNA:NTRK1} LMNA (NM_170707) - NTRK1 (NM_002529) fusion (LMNA exon 10 fused with NTRK1 exon 11) : c.1699-362:LMNA_c.1251+107:NTRK1del Protein Fusion: in frame {LMNA:NTRK1}	NA	
EVT-P-0023994-T06-IM6-29	P-0023994-T06-IM6		LMNA-NTRK1	LMNA	in-frame	three_prime	156844525	10	417	True	retained	1.0	False	Protein kinase domain				LMNA (NM_170707) - NTRK1 (NM_002529) fusion: c.1699-362:LMNA_c.1251+107:NTRK1del Protein Fusion: in frame {LMNA:NTRK1} LMNA (NM_170707) - NTRK1 (NM_002529) fusion (LMNA exon 10 fused with NTRK1 exon 11) : c.1699-362:LMNA_c.1251+107:NTRK1del Protein Fusion: in frame {LMNA:NTRK1}	NA	
EVT-P-0023994-T05-IM6-30	P-0023994-T05-IM6		LMNA-NTRK1	LMNA	in-frame	three_prime	156844525	10	417	True	retained	1.0	False	Protein kinase domain				LMNA (NM_170707) - NTRK1 (NM_002529) fusion: c.1699-362:LMNA_c.1251+107:NTRK1del Protein Fusion: in frame {LMNA:NTRK1} LMNA (NM_170707) - NTRK1 (NM_002529) fusion (LMNA exon 10 fused with NTRK1 exon 11) : c.1699-362:LMNA_c.1251+107:NTRK1del Protein Fusion: in frame {LMNA:NTRK1}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_retained	protein_lost	breakpoint_4	main_ref	donor_frame	main_ref	disrupted_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0023994-T01-IM6-31	P-0023994-T01-IM6		LMNA-NTRK1	LMNA	in-frame	three_prime	156844525	10	417	True	retained	1.0	False	Protein kinase domain				LMNA (NM_170707) - NTRK1 (NM_002529) fusion (LMNA exons 1-10 fused to NTRK1 exons 11-17): c.1699-362:LMNA_c.1251+107:NTRK1del	NA	
EVT-P-0025031-T02-IM6-32	P-0025031-T02-IM6		LTAP1-NTRK1	LTAP1	unknown	three_prime	156846272	14	571	False	disrupted	0.7814814814814814	False			Protein kinase domain		Fusion: in frame {LMNA:NTRK1} C1orf43 NA (NM_001098616) - NTRK1 (NM_002529): c.168+114:C1orf43_c.1713:NTRK1inv		
EVT-P-0024231-T01-IM6-33	P-0024231-T01-IM6		MEF2D-NTRK1	MEF2D	in-frame	five_prime	156845756	13	501	True	lost	0.0	False	Protein kinase domain				mid-exon {C1orf43:NTRK1} NTRK1 NA (NM_002529) - MEF2D (NM_005920) rearrangement: c.1502-116:NTRK1_c.1006+1908:MEF2Binv		
EVT-P-0045731-T01-IM6-35	P-0045731-T01-IM6		NOS1AP-NTRK1	NOS1AP	unknown	three_prime	156843455	8	294	False	retained	1.0	False	Protein kinase domain				Fusion: in frame {NTRK1:MEF2D} NOS1AP NA (NM_014697) - NTRK1 (NM_002529) fusion: c.1106-363:NOS1AP_c.881:NTRK1dup		
EVT-P-0026354-T01-IM6-36	P-0026354-T01-IM6		NTRK1-ARHGEF11	ARHGEF11	in-frame	three_prime	156845184	12	452	True	retained	1.0	False	Protein kinase domain				mid-exon {NOS1AP:NTRK1} ARHGEF11 NA (NM_198236) - NTRK1 (NM_002529) rearrangement: c.4631-285:ARHGEF11_c.1355-128:NTRK1inv		
EVT-P-0035466-T01-IM6-37	P-0035466-T01-IM6		NTRK1-ATP1A2	ATP1A2	unknown	three_prime	156844714	11	423	False	retained	1.0	False	Protein kinase domain				Fusion: in frame {ARHGEF11:NTRK1} ATP1A2 NA (NM_000702) - NTRK1 (NM_002529) rearrangement: c.2563+2:ATP1A2_c.1268:NTRK1dup		

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	donor_frame	inframe	disrupted_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0056197-T01-IM6-46	P-0056197-T01-IM6		NTRK1-NELFCD	NELFCD	out-of-frame	five_prime	156843148	8	284	True	lost	0.0	False			Protein kinase domain		NTRK1 (NM_002529) - NELFCD (NM_198976) rearrangement: t(1;20)(q23.1;q13.32)(chr1:g.156843148: Protein Fusion: out of frame {NTRK1:NELFCD}	NA (NM_001080471) - NTRK1 (NM_002529) fusion (PEAR1 exons 1-15 fused to NTRK1 exons 9-17): c.1951+466;PEAR1_c.1195+27:NTRK1	
EVT-P-0027277-T01-IM6-79	P-0027277-T01-IM6		NTRK1-PEAR1	PEAR1	in-frame	three_prime	156844219	9	399	True	retained	1.0	False			Protein kinase domain		PEAR1 (NM_001080471) - NTRK1 (NM_002529) fusion (PEAR1 exons 1-15 fused to NTRK1 exons 9-17): c.1951+466;PEAR1_c.1195+27:NTRK1		
EVT-P-0029892-T05-IM6-80	P-0029892-T05-IM6		NTRK1-PLEKHA6	PLEKHA6	in-frame	three_prime	156844239	9	399	True	retained	1.0	False			Protein kinase domain		PEAR1 (NM_001080471) - NTRK1 (NM_002529) fusion (PEAR1 exons 1-15 fused to NTRK1 exons 9-17): c.1951+466;PEAR1_c.1195+27:NTRK1		
EVT-P-0029892-T04-IM6-81	P-0029892-T04-IM6		NTRK1-PLEKHA6	PLEKHA6	in-frame	three_prime	156844239	9	399	True	retained	1.0	False			Protein kinase domain		PEAR1 (NM_001080471) - NTRK1 (NM_002529) fusion (PEAR1 exons 1-15 fused to NTRK1 exons 9-17): c.1951+466;PEAR1_c.1195+27:NTRK1		
EVT-P-0029892-T03-IM6-82	P-0029892-T03-IM6		NTRK1-PLEKHA6	PLEKHA6	in-frame	three_prime	156844239	9	399	True	retained	1.0	False			Protein kinase domain		PEAR1 (NM_001080471) - NTRK1 (NM_002529) fusion (PEAR1 exons 1-15 fused to NTRK1 exons 9-17): c.1951+466;PEAR1_c.1195+27:NTRK1		
EVT-P-0038819-T01-IM6-83	P-0038819-T01-IM6		NTRK1-RAB25	RAB25	out-of-frame	five_prime	156844458	10	417	True	lost	0.0	False			Protein kinase domain		PEAR1 (NM_001080471) - NTRK1 (NM_002529) fusion (PEAR1 exons 1-15 fused to NTRK1 exons 9-17): c.1951+466;PEAR1_c.1195+27:NTRK1		
EVT-P-0015004-T01-IM6-84	P-0015004-T01-IM6		NTRK1-RABGAP1L	RABGAP1L	out-of-frame	five_prime	156844972	11	452	True	lost	0.0	False			Protein kinase domain		PEAR1 (NM_001080471) - NTRK1 (NM_002529) fusion (PEAR1 exons 1-15 fused to NTRK1 exons 9-17): c.1951+466;PEAR1_c.1195+27:NTRK1		

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	is_main_ref	is_disrupted	is_disrupted	disrupted_desc	annotation	effect_on_fra
EVT-P-0038568-T01-IM6-85	P-0038568-T01-IM6		NTRK1-SEMA4A	SEMA4A	out-of-frame	five_prime	156841963	7	284	True	lost	0.0	False		Protein kinase domain		NTRK1 (NM_002529) - SEMA4A (NM_001193300) rearrangement: c.850+416:NTRK1_c.301-270:SEMA4A Protein Fusion: out of frame {NTRK1:SEMA4A}	NA
EVT-P-0048896-T01-IM6-86	P-0048896-T01-IM6		NTRK1-SMYD2	SMYD2	unknown	five_prime	156844382	10	405	False	lost	0.0	False		Protein kinase domain		NTRK1 (NM_002529) - SMYD2 (NM_020197) rearrangement: c.1215:NTRK1_c.602+876:SMYD2del Protein Fusion: mid-exon {NTRK1:SMYD2}	NA
EVT-P-0023987-T01-IM6-88	P-0023987-T01-IM6		NTRK1-TPR	TPR	unknown	three_prime	156844285	10	399	True	retained	1.0	False		Protein kinase domain		TPR (NM_003292) - NTRK1 (NM_002529) rearrangement: c.2902:TPR_c.1196-78:NTRK1inv Protein Fusion: mid-exon {TPR:NTRK1}	NA
EVT-P-0024549-T03-IM6-89	P-0024549-T03-IM6		NTRK1-TPR	TPR	in-frame	three_prime	156844290	10	399	True	retained	1.0	False		Protein kinase domain		TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2776+651_c.1196-73inv Protein Fusion: in frame {TPR:NTRK1}	NA
EVT-P-0007050-T01-IM5-90	P-0007050-T01-IM5		NTRK1-VANGL2	VANGL2	unknown	three_prime	156845141	12	452	True	retained	1.0	False		Protein kinase domain		VANGL2 (NM_020335) - NTRK1 (NM_002529) rearrangement : c.-191+4456:VANGL2_c.1355-171:NTRK1dup Transcript fusion (VANGL2-NTRK1)	NA
EVT-P-0006172-T01-IM5-91	P-0006172-T01-IM5		P2RY8-NTRK1	P2RY8	unknown	five_prime	156838012	5	182	False	lost	0.0	False		Protein kinase domain		P2RY8 (NM_008129) - NTRK1 (NM_002529) rearrangement: t(X;1)(chrX:g.1598513::chr1:g.156838012) Protein fusion: mid-exon (NTRK1-P2RY8)	NA
EVT-P-0042526-T02-IM6-92	P-0042526-T02-IM6		PRCC-NTRK1	PRCC	unknown	three_prime	156834591	3	120	False	retained	1.0	False		Protein kinase domain		PRCC (NM_005973) - NTRK1 (NM_002529) fusion: c.1323+10:PRCC_c.359:NTRK1del Protein Fusion: mid-exon {PRCC:NTRK1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	donor_frame	inframe	disrupted_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0069757-T01-IM7-100	P-0069757-T01-IM7		TARS2-NTRK1	TARS2	unknown	three_prime	156838028	5	187	False	retained	1.0	False	Protein kinase domain				TARS2 (NM_025150) - NTRK1 (NM_002529) fusion: c.264-188:TARS2_c.561:NTRK1del	NA	
EVT-P-0050767-T01-IM6-101	P-0050767-T01-IM6		TPM3-NTRK1	TPM3	in-frame	three_prime	156845252	12	452	True	retained	1.0	False	Protein kinase domain				Fusion: mid-exon {TARS2:NTRK1} TPM3(NM_158469) - NTRK1 (NM_002529) Fusion: TPM3 exon 12 with NTRK1 exon12): c.665-654:TPM3_c.1355-60:NTRK1inv	NA	
EVT-P-0049078-T04-IM6-102	P-0049078-T04-IM6		TPM3-NTRK1	TPM3	unknown	three_prime	156843537	8	321	False	retained	1.0	False	Protein kinase domain				Fusion: in frame {TPM3:NTRK1} TPM3 NA (NM_001278190) - NTRK1 (NM_002529) fusion: c.395-6179:TPM3_c.963:NTRK1inv	NA	
EVT-P-0049078-T01-IM6-103	P-0049078-T01-IM6		TPM3-NTRK1	TPM3	unknown	three_prime	156843537	8	321	False	retained	1.0	False	Protein kinase domain				Fusion: mid-exon {TPM3:NTRK1} TPM3 NA (NM_001278190) - NTRK1 (NM_002529) fusion: c.395-6179:TPM3_c.963:NTRK1inv	NA	
EVT-P-0046877-T02-IM6-104	P-0046877-T02-IM6		TPM3-NTRK1	TPM3	in-frame	three_prime	156843811	8	393	True	retained	1.0	False	Protein kinase domain				Fusion: mid-exon {TPM3:NTRK1} TPM3 NA (NM_152263) - NTRK1 (NM_002529) fusion: c.776-357:TPM3_c.1177+60:NTRK1inv	NA	
EVT-P-0026375-T04-IM6-105	P-0026375-T04-IM6		TPM3-NTRK1	TPM3	in-frame	three_prime	156844046	9	393	True	retained	1.0	False	Protein kinase domain				Fusion: in frame {TPM3:NTRK1} TPM3 NA (NM_152263) - NTRK1 (NM_002529) fusion: c.775+355:TPM3_c.1178-129:NTRK1inv	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint	breakpoint	breakpoint	protein	protein	breakpoint	main_ref	donor	acceptor	disrupted_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0026375-T01-IM6-106	P-0026375-T01-IM6		TPM3-NTRK1	TPM3	in-frame	three_prime	156844046	9	393	True	retained	1.0	False	Protein kinase domain				TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-8 fused to NTRK1 exons 9-17): c.775+355:TPM3_c.1178-129:NTRK1inv Protein Fusion: in frame {TPM3:NTRK1} TPM3 NA (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-6 with NTRK1 exons 10-17): c.*6032:TPM3_c.1195+38:NTRK1inv Protein Fusion: in frame {TPM3:NTRK1} TPM3 NA (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-6 fused in-frame to NTRK1 exons 9-17): c.*652:TPM3_c.1178-137:NTRK1inv Protein Fusion: in frame {TPM3:NTRK1} TPM3 NA (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-8 fused to NTRK1 exons 12-17): c.775+235:TPM3_c.1355-94:NTRK1inv Protein Fusion: in frame {TPM3:NTRK1}	NA	
EVT-P-0025815-T02-IM6-107	P-0025815-T02-IM6		TPM3-NTRK1	TPM3	in-frame	three_prime	156844230	9	399	True	retained	1.0	False	Protein kinase domain				TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-6 with NTRK1 exons 10-17): c.*6032:TPM3_c.1195+38:NTRK1inv Protein Fusion: in frame {TPM3:NTRK1} TPM3 NA (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-6 fused in-frame to NTRK1 exons 9-17): c.*652:TPM3_c.1178-137:NTRK1inv Protein Fusion: in frame {TPM3:NTRK1} TPM3 NA (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-8 fused to NTRK1 exons 12-17): c.775+235:TPM3_c.1355-94:NTRK1inv Protein Fusion: in frame {TPM3:NTRK1}	NA	
EVT-P-0021467-T01-IM6-108	P-0021467-T01-IM6		TPM3-NTRK1	TPM3	in-frame	three_prime	156844038	9	393	True	retained	1.0	False	Protein kinase domain				TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-6 fused in-frame to NTRK1 exons 9-17): c.*652:TPM3_c.1178-137:NTRK1inv Protein Fusion: in frame {TPM3:NTRK1} TPM3 NA (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-8 fused to NTRK1 exons 12-17): c.775+235:TPM3_c.1355-94:NTRK1inv Protein Fusion: in frame {TPM3:NTRK1}	NA	
EVT-P-0021170-T03-IM6-109	P-0021170-T03-IM6		TPM3-NTRK1	TPM3	in-frame	three_prime	156845218	12	452	True	retained	1.0	False	Protein kinase domain				TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-8 fused to NTRK1 exons 12-17): c.775+235:TPM3_c.1355-94:NTRK1inv Protein Fusion: in frame {TPM3:NTRK1}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	inframe	main_ref	disrupted	description	annotation	effect_on_fra
EVT-P-0021170-T02-IM6-110	P-0021170-T02-IM6		TPM3-NTRK1	TPM3	in-frame	three_prime	156845218	12	452	True	retained	1.0	False	Protein kinase domain		TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-8 fused to NTRK1 exons 12-17): c.775+235:TPM3_c.1355-94:NTRK1inv Protein Fusion: in frame {TPM3:NTRK1} TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-10 fused in-frame to NTRK1 exons 9-17): c.*1568:TPM3_c.1178-173:NTRK1inv Protein fusion: in frame {TPM3:NTRK1} TPM3 (NM_152263) - NTRK1 (NM_002529) fusion: c.854+282:TPM3_c.1027:NTRK1inv Protein Fusion: mid-exon {TPM3:NTRK1} TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2610+187:TPR_c.1252-38:NTRK1inv Protein Fusion: in frame {TPR:NTRK1} TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2776+531:TPR_c.1129:NTRK1inv Protein Fusion: mid-exon {TPR:NTRK1} TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2777-512:TPR_c.1077:NTRK1inv Protein Fusion: mid-exon {TPR:NTRK1}	NA	
EVT-P-0014143-T01-IM5-111	P-0014143-T01-IM5		TPM3-NTRK1	TPM3	in-frame	three_prime	156844002	9	393	True	retained	1.0	False	Protein kinase domain		TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-10 fused in-frame to NTRK1 exons 9-17): c.*1568:TPM3_c.1178-173:NTRK1inv Protein fusion: in frame {TPM3:NTRK1} TPM3 (NM_152263) - NTRK1 (NM_002529) fusion: c.854+282:TPM3_c.1027:NTRK1inv Protein Fusion: mid-exon {TPM3:NTRK1} TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2610+187:TPR_c.1252-38:NTRK1inv Protein Fusion: in frame {TPR:NTRK1} TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2776+531:TPR_c.1129:NTRK1inv Protein Fusion: mid-exon {TPR:NTRK1} TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2777-512:TPR_c.1077:NTRK1inv Protein Fusion: mid-exon {TPR:NTRK1}	NA	
EVT-P-0069266-T01-IM7-112	P-0069266-T01-IM7		TPM3-NTRK1	TPM3	unknown	three_prime	156843601	8	343	False	retained	1.0	False	Protein kinase domain		TPM3 (NM_152263) - NTRK1 (NM_002529) fusion: c.854+282:TPM3_c.1027:NTRK1inv Protein Fusion: mid-exon {TPM3:NTRK1} TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2610+187:TPR_c.1252-38:NTRK1inv Protein Fusion: in frame {TPR:NTRK1} TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2776+531:TPR_c.1129:NTRK1inv Protein Fusion: mid-exon {TPR:NTRK1} TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2777-512:TPR_c.1077:NTRK1inv Protein Fusion: mid-exon {TPR:NTRK1}	NA	
EVT-P-0049951-T01-IM6-113	P-0049951-T01-IM6		TPR-NTRK1	TPR	in-frame	three_prime	156844660	11	418	True	retained	1.0	False	Protein kinase domain		TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2610+187:TPR_c.1252-38:NTRK1inv Protein Fusion: in frame {TPR:NTRK1} TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2776+531:TPR_c.1129:NTRK1inv Protein Fusion: mid-exon {TPR:NTRK1} TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2777-512:TPR_c.1077:NTRK1inv Protein Fusion: mid-exon {TPR:NTRK1}	NA	
EVT-P-0042965-T01-IM6-114	P-0042965-T01-IM6		TPR-NTRK1	TPR	unknown	three_prime	156843703	8	377	False	retained	1.0	False	Protein kinase domain		TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2610+187:TPR_c.1252-38:NTRK1inv Protein Fusion: in frame {TPR:NTRK1} TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2776+531:TPR_c.1129:NTRK1inv Protein Fusion: mid-exon {TPR:NTRK1} TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2777-512:TPR_c.1077:NTRK1inv Protein Fusion: mid-exon {TPR:NTRK1}	NA	
EVT-P-0065034-T02-IM7-115	P-0065034-T02-IM7		TPR-NTRK1	TPR	unknown	three_prime	156843651	8	359	False	retained	1.0	False	Protein kinase domain		TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2610+187:TPR_c.1252-38:NTRK1inv Protein Fusion: in frame {TPR:NTRK1} TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2776+531:TPR_c.1129:NTRK1inv Protein Fusion: mid-exon {TPR:NTRK1} TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2777-512:TPR_c.1077:NTRK1inv Protein Fusion: mid-exon {TPR:NTRK1}	NA	

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3 Figures

NTRK1 domain-retention track

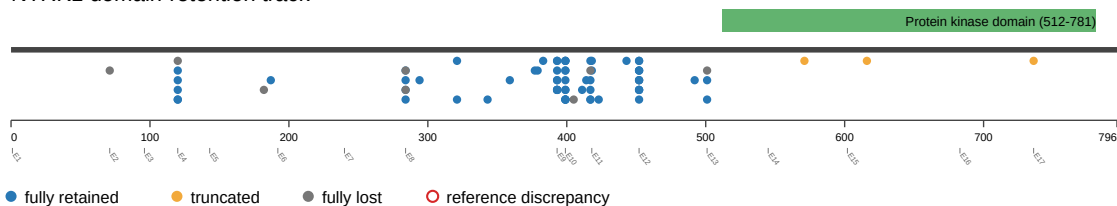
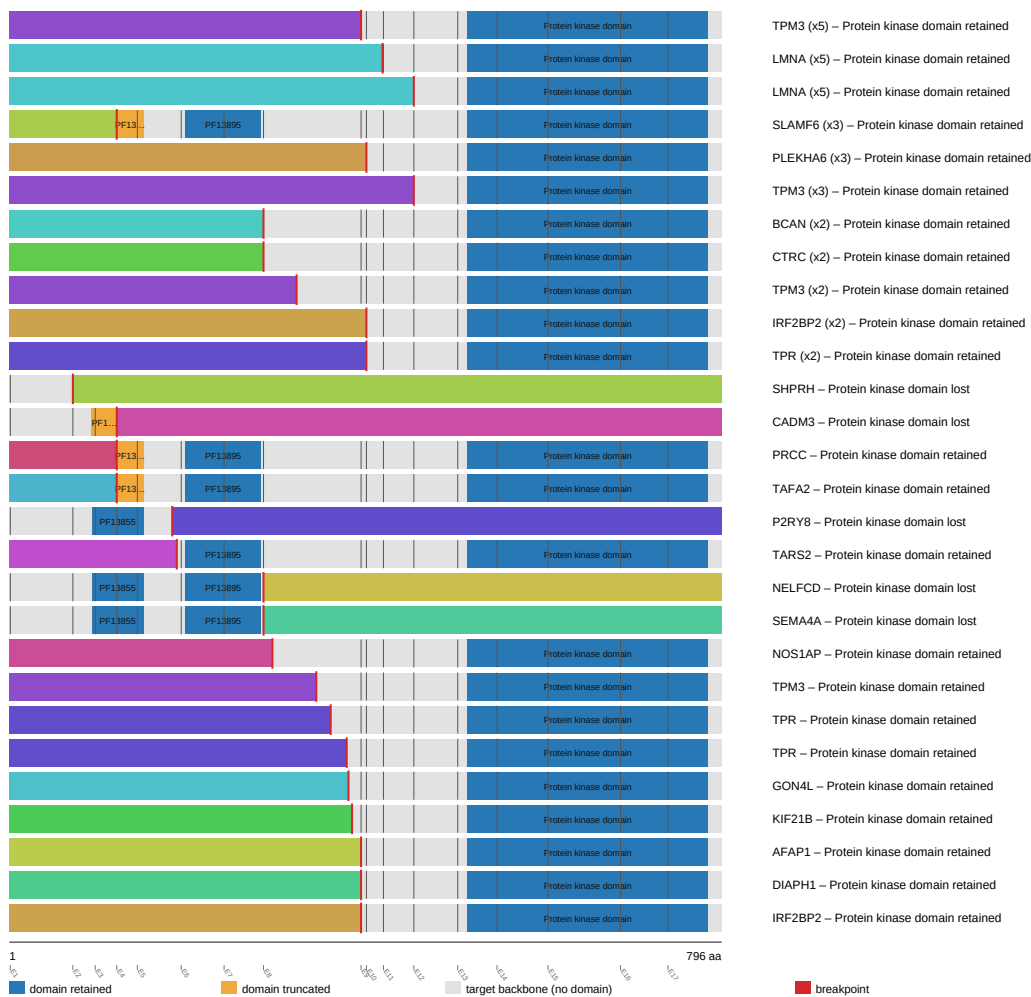


Figure 1: domain retention outliers

NTRK1 fusion-transcript schematic

partner block → breakpoint → retained NTRK1 portion (domain-colored, exon ticks)



Showing the top 28 of 53 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

Figure 2: fusion schematic

NTRK1 intragenic-deletion schematic

retained N-terminal block → deleted span (plain connector) → resumed C-terminal block

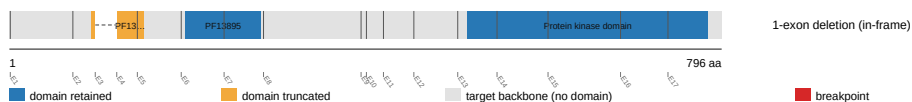


Figure 3: intragenic deletion schematic

Reference vs msk_impact_50k_2026



Figure 4: reference comparison